



Module 3 challenge

1. While using **VLOOKUP**, you encounter an error because some of your spreadsheet values have leading and trailing spaces. What function should you use to eliminate these spaces?

1 / 1 point

- ☒ **TRIM**
- ☐ **CUT**
- ☐ **NOSPACE**
- ☐ **VALUE**

✓ Correct

2. Fill in the blank: The spreadsheet function _____ can be used to add up the number of times a range of cells contains the value "paid."

1 / 1 point

- ☐ **COUNT DISTINCT**
- ☐ **RANGE**
- ☐ **RETURN**
- ☒ **COUNT**

✓ Correct

3. A data analyst writes the following formula: =MAX(\$E\$5:\$E\$500). What are the purposes of the dollar signs (\$) ? Select all that apply.

1 / 1 point

☒ Create an absolute reference.

✔ Correct

☒ Find the maximum value in the range E5 to E500 regardless of whether the formula is copied.

✔ Correct

☐ Perform the calculation more efficiently.

☒ Ensure rows and columns do not change.

✔ Correct

4. What will this query return?

1 / 1 point

```
SELECT *  
FROM Inventory_table  
LEFT JOIN Scrap_table
```

- ☒ All records in **Inventory_table** and any matching rows from **Scrap_table**
- ☐ All records in both **Inventory_table** and **Scrap_table**
- ☐ All records in **Scrap_table** and any matching rows from **Inventory_table**
- ☐ All rows from **Inventory_table** joined together with **Scrap_table**

✔ Correct

5. In this spreadsheet, which function will search for the water type of Lake Urmia?

1 / 1 point

	A	B	C
1	Lake	Surface area (sq. miles)	Water type
2	Caspian Sea	143,000	Saline
3	Superior	31,700	Freshwater
4	Victoria	26,590	Freshwater
5	Huron	23,000	Freshwater
7	Tanganyika	12,600	Freshwater
8	Balkhash	6,300	Saline
9	Athabasca	3,030	Freshwater
10	Urmia	2,320	Freshwater

- ☐ `=VLOOKUP("Urmia", B2:C10, 2, false)`
- ☐ `=VLOOKUP(Urmia, A2:C10, 2, false)`
- ☒ `=VLOOKUP("Urmia", A2:C10, 3, false)`
- ☐ `=VLOOKUP(Urmia, A2:B10, false)`

✔ Correct

6. Fill in the blank: A SQL clause containing **HAVING** adds a _____ to a query instead of the underlying table.

1 / 1 point

- ☐ join
- ☒ filter
- ☐ subquery
- ☐ limit

✔ Correct

7. A data professional at a manufacturing company works with a spreadsheet containing inventory management data. To create a chart from the data to share with stakeholders, the analyst uses a function that converts all data to numeric values. What function do they use?

1 / 1 point

- ☐ NUM
- ☒ VALUE
- ☐ EXCHANGE
- ☐ PROCESS

✓ Correct

8. Which query will select all columns from the **highways** table and alias the table to **hwys**?

1 / 1 point

☐ SELECT *
FROM highways NEW hwys

☒ SELECT *
FROM highways AS hwys

☐ SELECT *
FROM highways ALIAS hwys

☐ SELECT *
FROM highways TO hwys

✓ Correct

9. What are the benefits of using subqueries in SQL? Select all that apply.

0.5 / 1 point

☒ Subqueries can replace complex joins in some cases.

✔ Correct

☐ Subqueries can improve code readability.

☒ Subqueries can be used to filter data based on results from another table.

✘ This should not be selected

Review [the reading about SQL functions and subqueries.](#) [↗](#)

☒ Subqueries can simplify complex queries.

✔ Correct

10. What does this code do?

1 / 1 point

```
1 SELECT account_table.*
2 FROM (
3     SELECT *
4     FROM transaction.sf_model_feature_2014_01
5     WHERE day_of_week = 'Monday'
6 ) AS account_table
7 WHERE account_table.availability = 'NO'
```

- ☐ It selects all data from the transaction.sf_model_feature_2014_01 table.
- ☐ It filters the transaction.sf_model_feature_2014_01 table to include only transactions on Fridays.
- ☐ It creates a temporary table named account_table containing all transactions from transaction.sf_model_feature_2014_01.
- ☒ It selects all columns from the table containing transactions on Monday and filters them to display only the transactions that are unavailable.

✔ Correct

11. What does this code do?

1 / 1 point

```
1  SELECT e.employee_id, e.first_name, e.last_name, e.salary
2  FROM employees e
3  WHERE e.salary > (
4      SELECT AVG(salary)
5      FROM employees
6      WHERE department_id = e.department_id
7  );
8
```

- ☒ It finds employees who earn more than the average salary in their department.
- ☐ It calculates the average salary for all employees.
- ☐ It groups employees by department.
- ☐ It joins the employees table with itself.

✓ Correct